

The analytic two-dimensional cutoff theorem

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Abstract

For the equal-exponent block and an amplitude given on the box by a double power series with a radius margin, $\Phi(u, v, s) = \sum c_{ij}(s) u^i v^j$ with $|c_{ij}(s)| \rho^{i+j} \leq H(s) \leq C_0(1+s)^D e^{\beta s L}$ and $b < \rho$, the reduced cutoff expansion of units 107–109 holds with the faces $a_i = \sum_j c_{ij} v^j$, $b_j = \sum_i c_{ij} u^i$ and the corner jets c_{ij} , and no further hypotheses:

$$Z_\Phi(N) - \text{reducedSum}(N) = O(N^{-(p+\min(M_1/k_1, M_2/k_2))}(1 + \log N)).$$

All face remainders, the mixed remainder and the moment hypotheses are discharged by units 110 and 114. This is the paper-facing higher-order $d = 2$ theorem under the paper's own analyticity assumption, in the Schwarz-free form recommended in Astra #6. Zero `sorry/axiom`.

1 The things formalised

```
theorem uface_gamma_lt (p : ) (h h k k M M i : ) (hk : 0 < k) (hk : 0 < k)
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (hM : ((M : ) - 1) / k < (M : ) / k) (hi : i < M) :
  (k : ) * (((h + i : ) : ) + 1) / k - h - 1 < M
theorem vface_gamma_lt / corner_gamma_lt -- the other two resonance parameters
theorem anaCoeff_env ... (hm : m < M) (s : ) (hs : 0 < s) :
  |c (i, m) s| C * ( ^ i ) ^ 1 * (1 + ^ 1) ^ M * (1 + s) ^ D * Real.exp ( * s * L)
theorem twoD_cutoff_analytic ( b p C L : ) (h h k k M M D : ) (h : 0 < )
  (hb : 0 < b) (hb : b < ) (hk : 0 < k) (hk : 0 < k) (hp : ((h : ) + 1) / k = p)
  (hp : ((h : ) + 1) / k = p)
  (hM : ((M : ) - 1) / k < (M : ) / k) (hM : ((M : ) - 1) / k < (M : ) / k)
  (c : × → → ) (hcc : ij, Continuous (c ij)) (H : → ) (hH : Continuous H)
  (hc : ij s, |c ij s| * ^ (ij.1 + ij.2) H s) (hC : 0 C)
  (henv : s, 0 s → H s C * (1 + s) ^ D * Real.exp ( * s * L)) :
  (fun N : => twoDAMP b N h h k k (anaAmp c b)
    - reducedSum b h h k k M M (anaFaceU c b) (anaFaceV c b)
    (fun i j s => c (i, j) s) N)
  =0[atTop] fun N : =>
    N ^ (-(p + min ((M : ) / k) ((M : ) / k))) * (1 + Real.log N)
```

2 Mathematics

The theorem is an instantiation of `twoD_cutoff_reduced_isBig0`. Continuity of the amplitude and faces is unit 114's; the face remainders are $H(s)\rho^{-i-M_2}/(1-b/\rho) \cdot v^{M_2}$ and the mixed remainder $H(s)\rho^{-M_1-M_2}/(1-b/\rho)^2 \cdot u^{M_1}v^{M_2}$, both of the required envelope form once $H \leq C_0(1+s)^D e^{\beta s L}$. The `FaceMoments` bundles come from unit 110's constructor: every coefficient c_{im} with $m < M$

is bounded by $C_0 \rho^{-i} (1 + \rho^{-1})^M (1 + s)^D e^{\beta s L}$, the face envelopes by the corresponding constants, and the corner data by $C_0 \rho^{-i-j}$ with a zero remainder. The resonance conditions $\gamma < M$ for the u -faces, v -faces and corners are the same arithmetic as in unit 107, now extracted as lemmas.

3 Paper-facing meaning

Under the paper's standing hypothesis (analytic data near the exceptional divisor, here in the form of a convergent double series on a larger polydisc), the $d = 2$ equal-exponent block has the complete finite-cutoff expansion: exponents $p + i/k_1$ and $p + j/k_2$, logarithms exactly at collisions, constant coefficients given by the weighted finite parts of the face series, remainder $O(N^{-(p+\tau)}(1 + \log N))$ with $\tau = \min(M_1/k_1, M_2/k_2)$. No compatibility or moment hypothesis remains. What the theorem does not yet do: produce the double series of $\eta e^{\beta s \xi}$ from those of ξ, η (the weighted- ℓ^1 coefficient closure, Astra #6 rank 2) and merge the three log families into one copy per collision.

4 Lean notes

- Symmetric hypotheses are discharged by symmetric lemmas: `vface_gamma_lt` is `uface_gamma_lt` with the coordinates swapped, and the v -face envelopes come from `anaCoeff_env` applied to `swapC c` followed by `simp [swapC]`.
- A two-factor inequality inside `calc` is `mul_le_mul` with two `positivity` side goals; `gcongr` may close the main goal and leave a side goal in an unexpected slot.
- Long theorem applications with many `?_` placeholders are more legible as `refine ... ?_ ?_` followed by bullet proofs than as inline lambdas.